

LEO ZHOU

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Education

Rice University | B.S. in Computer Science | GPA: 3.92/4.00 Expected Graduation May 2028
• **Relevant Coursework:** Data Structures & Algorithms, Systems Programming, Discrete Math, Linear Algebra

Experience

Undergraduate Research Assistant | Rice University | Prof. Jiarong Xing August 2026 – Present
• Ended a forced daily re-login in a macOS usage-tracking app by implementing OAuth token refresh in Swift
• Wrote a Keychain credential inspector after the documented token shape proved incomplete, verified live
• Surfaced usage from Claude Code sessions run over SSH by discovering hosts and mirroring transcripts

Software Engineer Intern | CanSemi June 2026 – August 2026
• Built an event-driven Python parser pulling measurements from 500+ binary test equipment (ATE) logs
• Designed a normalized PostgreSQL schema and loader ingesting 40 wafer lots (~12,000 records) in under 2 min
• Wrote a FastAPI service with 8 endpoints over that data, cutting report turnaround ~75% for 6 engineers
• Built an interactive React and Canvas wafer map with hover and drill-down, retiring the team's Excel heatmaps

Software Engineer | Inkstone Technologies April 2026 – July 2026
• Built a native iOS reading app in Swift and SwiftUI that adapts every screen to the user's known characters
• Wired Apple Vision OCR into a live camera pipeline overlaying pronunciation on unfamiliar characters only
• Parsed and frequency-ranked a 113,000-entry Chinese dictionary in Python into an indexed SQLite database
• Shipped handwriting input on ML Kit Digital Ink, StoreKit subscriptions, and a WidgetKit home screen widget

Embedded Software Engineer | Rice Wind Energy October 2025 – April 2026
• Programmed ESP32 firmware in C/C++ reading turbine speed, power output, and blade pitch off live sensors
• Built a six-state safety machine with fault latching and recovery, plus perturb-and-observe power tracking

Projects

BlunderNet | Live Chess Training Platform 🌐 | 🌐 blundernet.com 2026
Go, AWS, Terraform, Docker, Redis, PostgreSQL, React, PyTorch, C++, Stockfish
• Built and ran a live chess site for ~120 users, filtering 3.25M puzzles by rating, theme, phase and length
• Cut uniform random puzzle selection from 1.4 s to 0.9 ms with a precomputed grid and a stored shuffle key
• Kept the Go API stateless with game state in Redis, running engine moves and Stockfish reviews on SQS workers
• Built the React front end with a screen-readable board and blindfold mode, live over WebSockets and pub/sub
• Trained the engine in PyTorch from scratch, rewriting its Monte Carlo tree search in C++ for a 1.8× speedup
• Defined the AWS stack in Terraform and load-tested with k6, watching workers autoscale 1 to 4 under backlog

Vestigo | Evidence-Grounded Geolocation Agent 🌐 | 🌐 vestigo.earth Ongoing
Python, PyTorch, CLIP/SigLIP, LLM agent design, tool calling, evaluation harness
• Building a photo geolocation agent: a five-step loop of observe, guess, call tools, claim, then resolve
• Cut classifier median error 1,024 km to 142 km, within 1.5x of a frontier model call on the same photos
• Traced the ceiling to the frozen encoder, not the data: three swaps beat tripling the training set
• Held calibration at 3.7% after temperature scaling, so a more accurate model stayed an honest one

Technical Skills

Languages: Python, Go, C/C++, Swift, TypeScript/JavaScript, SQL, HTML/CSS
Backend & Web: FastAPI, React, SwiftUI, REST APIs, WebSockets, PostgreSQL, Redis, SQLite
Cloud & Infrastructure: AWS (ECS, SQS, ALB, RDS), Terraform, Docker, Linux, SSH, GitHub Actions, Git, k6
ML & AI: PyTorch, CLIP, SigLIP, ONNX, LLM agents, model calibration, Apple Vision, ML Kit
Embedded: ESP32, STM32 (bare-metal), C/C++ firmware, state machines, UART/SPI/I²C

Leadership & Athletics

- **New Zealand Junior National Fencing Team** — Captain, Coach, Referee; 30+ titles, Junior Worlds
- **Rice Fencing Club** — Manager · **Physics Pathfinders** — Charity President; 600+ students, 22 schools